

# Numerical Methods

## Chapter-2

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# Algebraic and Transcendental Equations

**Algebraic Function:** An algebraic function is a function that can be defined as the root of a polynomial equation. Example-  $f(x)=x^2 + x - 6$ ,  $f(x)=2x^3 + 4x^2 - x + 7$  etc.

**Algebraic Equation:** An equation in the form of a polynomial having a finite number of terms and equated to zero is called algebraic equation. Example-  $x^2 + x - 6 = 0$ ,  $2x^3 + 4x^2 - x + 7 = 0$  etc.

# Algebraic and Transcendental Equations(Cont'd)

**Transcendental Function:** A transcendental function is an analytic function that does not satisfy a polynomial equation, in contrast to an algebraic function. Example-  $f(x)=\ln x^3 - 0.7$ ,  $\varphi(x)=e^{-0.5x} - 5x$ ,  $\psi(x)=\sin^2 x - x^2 - 2$  etc.

**Transcendental Equation:** A transcendental equation is an equation containing a transcendental function of the variable(s) being solved for. Example-  $x = \frac{1}{5} e^{-0.5x}$ ,  $x^2 = \sin^2 x - 2$  etc.

# Algebraic and Transcendental Equations(Cont'd)

If  $f(x)$  is a polynomial of the form  $f(x)=a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$  then we can say from the theory of equations for locating its roots-

1. Every polynomial equation of the  $n$ th degree has  $n$  and only  $n$  roots.
2. If  $n$  is odd, the polynomial equation has at least one real root whose sign is opposite to that of the last term.

# Algebraic and Transcendental Equations(Cont'd)

3. If  $n$  is even and the constant term is negative, then the equation has at least one positive root and at least one negative root.

# Roots of an Equation

**Roots of an Equation:** Roots of the equation are such values of the variable that turn equation into correct equality. For example- 3 is the root of the equation  $15 = x^2 + 2x$ .

Roots of the equation is calculated by the following methods-

- ❑ Bisection method
- ❑ False Position / Regula-Falsi method / method of chords
- ❑ Iteration method
- ❑ Newton Raphson method
- ❑ Ramanujan's method, etc

# Roots of an Equation (Cont'd)

**Bracketing Methods:** These methods require two initial guesses,  $a$  and  $b$ , such that the root is known to lie between them. This is typically achieved when  $f(a)$  and  $f(b)$  have opposite signs (i.e.,  $f(a) \cdot f(b) < 0$ ), ensuring a sign change and, for continuous functions, the presence of at least one root within the interval  $[a, b]$ .

**Examples:** Bisection Method, False Position Method.

**Advantages:** Guaranteed convergence to a root if one exists within the initial bracket.

**Disadvantages:** Generally slower convergence compared to open methods.

# Roots of an Equation (Cont'd)

**Open Methods:** These methods do not require the initial guesses to bracket the root. They use one or more initial guesses to generate a sequence of approximations that ideally converge to the root.

**Examples:** Newton-Raphson Method, Secant Method, Fixed-Point Iteration.

**Advantages:** Can converge much faster than bracketing methods if the initial guess is sufficiently close to the root and the function behaves well.

**Disadvantages:** Convergence is not guaranteed and can diverge if the initial guess is poor or the function has certain characteristics.

# Bisection Method

If a function  $f(x)$  is continuous between  $a$  and  $b$ , and  $f(a)$  and  $f(b)$  are of opposite signs, then there exists at least one root between  $a$  and  $b$ . Suppose  $f(a)$  be negative and  $f(b)$  be positive then the root lies between  $a$  and  $b$  and its approximate value be given by  $x_0 = (a + b)/2$ . If  $f(x_0) = 0$ , we conclude that  $x_0$  is a root of the equation  $f(x) = 0$ . Otherwise, the root lies either between  $x_0$  and  $b$ , or between  $x_0$  and  $a$  depending on whether  $f(x_0)$  is negative or positive. If there are more roots than one in the interval, bisection method finds one of the roots.

# Bisection Method(Cont'd)

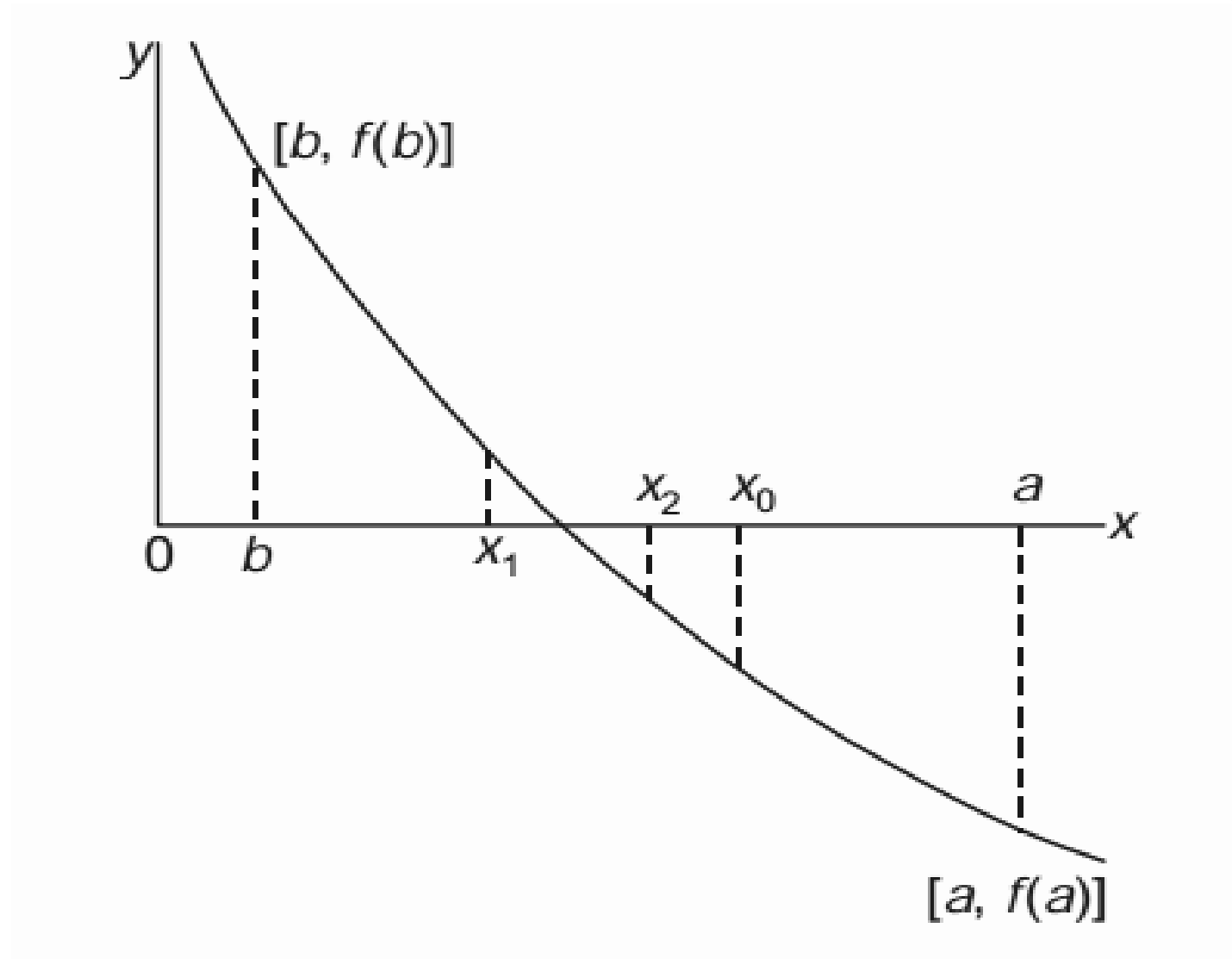


Fig: Graphical representation of the bisection method.

# Bisection Method(Cont'd)

It can be easily programmed using the following computational steps:

1. Choose two real numbers  $a$  and  $b$  such that  $f(a)f(b) < 0$ .
2. Set  $x_r = (a + b)/2$ .
3. (a) If  $f(a)f(x_r) < 0$ , the root lies in the interval  $(a, x_r)$ . Then, set  $b = x_r$  and go to step 2 above.  
  
(b) If  $f(a)f(x_r) > 0$ , the root lies in the interval  $(x_r, b)$ . Then, set  $a = x_r$  and go to step 2 above.  
  
(c) If  $f(a)f(x_r) = 0$ , it means that  $x_r$  is a root of the equation  $f(x) = 0$  and the computation may be terminated.

# Bisection Method(Cont'd)

We have to keep in mind that in practical problems, the roots may not be exact so that condition (c) in previous page is never satisfied. In such a case, we need to adopt a criterion for deciding when to terminate the computations. A convenient criterion is to compute the percentage error,  $\varepsilon_r$  defined by-

$$\varepsilon_r = \left| \frac{x'_r - x_r}{x'_r} \right| \times 100 \%$$

Where  $x'_r$  is the new value of  $x_r$ . The computations can be terminated when  $\varepsilon_r$  becomes less than a prescribed tolerance say  $\varepsilon_p$ . In addition, the maximum number of iterations may also be specified in advance.

# Bisection Method(Cont'd)

**Example:** Find a real root of the equation  $x^3 - 2x - 5 = 0$  correct to three decimal places.

**Solution:** Given that,

$$x^3 - 2x - 5 = 0$$

Let,  $f(x) = x^3 - 2x - 5$

Taking  $a=2$  and  $b=3$

$$\therefore f(2) = 2^3 - 2 \times 2 - 5 = -1 (< 0)$$

$$\therefore f(3) = 3^3 - 2 \times 3 - 5 = 16 (> 0)$$

Hence a root lies between 2 and 3.

$$\therefore x_r = \frac{a+b}{2} = \frac{2+3}{2} = 2.5$$

# Bisection Method(Cont'd)

| Step No. | a       | b         | $x_r = \frac{a+b}{2}$ | $f(x) = x^3 - 2x - 5$ |
|----------|---------|-----------|-----------------------|-----------------------|
| 1        | 2       | 3         | 2.5                   | 5.625                 |
| 2        | 2       | 2.5       | 2.25                  | 1.8906                |
| 3        | 2       | 2.25      | 2.125                 | 0.34570               |
| 4        | 2       | 2.125     | 2.0625                | -0.3513               |
| 5        | 2.0625  | 2.125     | 2.09375               | -0.00894              |
| 6        | 2.09375 | 2.125     | 2.109375              | 0.16683               |
| 7        | 2.09375 | 2.109375  | 2.1015625             | 0.07856               |
| 8        | 2.09375 | 2.1015625 | 2.09765               | 0.03471               |
| 9        | 2.09375 | 2.09765   | 2.0957                | 0.01282               |
| 10       | 2.09375 | 2.0957    | 2.094725              | 0.001936              |
| 11       | 2.09375 | 2.094725  | 2.0942375             | -0.0035               |

# Bisection Method(Cont'd)

The difference between  $x_{10}$  and  $x_{11}$  is 0.0004875. Hence a root, correct to three decimal places is 2.094.

## Self Practise

**Example-** 2.1, 2.3, 2.4, 2.5

**N.B.:** While computing trigonometric ratio, the specified value must be converted into degree using,

$$\text{Degree} = \frac{180 \times X}{3.14}$$

Where, X is in radian unit.

# Task related to Bisection Method

1. Write some advantages of bisection method.
2. Write some disadvantages of bisection method.

***Any Question???***